IBM Z DATATHON 2026

Z-MIRROR

Production-shaped test data. Released with evidence.

A synthetic-data pipeline for COBOL-oriented testing that combines reproducible generation, format validation, measured utility and an explicit release decision.

LINUXONE | s390x | PYTHON + DOCKER

THE DIFFERENCE

Not just synthetic rows.
A trail of evidence.

Current status is disclosed: generator + validator tests pass; utility, privacy-risk and COBOL execution gates remain open.

The testing-data gap

Useful test data needs more than valid-looking values.

01

Production copies

Moving real customer records into development and QA can create avoidable exposure.

02

Toy synthetic data

Unfitted random data can lose relationships and mislead downstream tests.

03

No release evidence

Generation alone does not prove structure, utility or privacy safety.

Z-MIRROR RESPONSE

Generate → validate → measure → assess risk → report a release decision

Target users: COBOL developers, QA engineers, data engineers, data stewards and mainframe modernisation teams.

How the pipeline works

A small, inspectable path from a public reference to an evidence-backed decision.

01

Reference

Public UCI data
45,211 rows

02

Generate

Seeded candidate
1,000 rows

03

Validate

Schema, ranges,
unique IDs, encoding

04

Evaluate

Utility + empirical
privacy checks

05

Release gate

PASS / FAIL /
INCONCLUSIVE

IMPLEMENTED / DEMONSTRATED

Generation · manifest · schema/range validation · regression tests

REMAINING RELEASE GATES

V2 utility thresholds · privacy-risk checks · actual COBOL run

No automatic anonymity claim: each gate must be backed by observed results and explicit limitations.

Built and exercised on IBM LinuxONE

The current deployment proves execution on s390x; application-level COBOL compatibility is still pending.

REFERENCE INPUT

UCI Bank Marketing

LINUXONE COMMUNITY CLOUD · s390x

Docker container · Python · JupyterLab

Seeded generator + validator + manifests

ARTIFACTS

CSV · hash manifest · reports

OUTPUT / VALIDATION

Candidate CSV → 12-character fixed-width sample; the five-record fixture was measured at 12 characters per line.

COBOL EXECUTION

GnuCOBOL 3.1.2 is installed on the host; compiling and running a real consumer is not yet evidenced.

Data boundary for this prototype: public benchmark data only. No customer dataset is used in the demonstrated run.

What is verified so far

These are observed prototype results, not claims of production readiness.

8 / 8

Generator tests

Reproducibility, seeds, schema, row counts, valid ranges, manifest hash.

18 / 18

Validator scenarios

Invalid headers/fields, IDs, value bounds, malformed CSV, stale artifacts.

1,000

V2 candidate rows

Notebook reports 124 aggregate cells, 45,211 reference rows and seed 42.

IMPORTANT FINDING

An earlier V1 comparison had a 32.2 percentage-point outcome-rate gap (reference 11.7%, V1 candidate 43.9%). This is why V2 needs a fresh measured comparison.

V2 utility results and policy thresholds are not yet confirmed. Do not report the V1 metrics as V2 results.

Release gate: honest status, not a rubber stamp

Z-Mirror is currently a testable prototype, not a privacy certification system.

Generator regression suite	PASS	8/8 tests passed independently
Extended validator suite	PASS	18/18 test scenarios passed independently
Generator V2 utility	INCONCLUSIVE	Fresh V2/reference comparison + thresholds still needed
Empirical privacy-risk checks	NOT VERIFIED	Implement risk tests; no formal anonymity / DP claim
COBOL consumer execution	NOT VERIFIED	Compiler installed; actual compile/run still required

Release rule: a failing or unevaluated gate must not be converted to PASS merely to improve a demo.

Who benefits — and what comes next

A small evidence workflow can make test-data decisions more explainable for mainframe teams.

WHO BENEFITS

QA / test engineers: repeatable fixture generation.

COBOL developers: layout checks and planned consumer tests.

Data stewards: a visible record of checks, gaps and limitations.

NEXT VALIDATION STEPS

1. Measure Generator V2 utility against the public reference.
2. Add empirical privacy-risk checks and report limitations.
3. Compile/run a real GnuCOBOL consumer on the fixed-width file.

OPEN-SOURCE RESOURCES

UCI Bank Marketing · Python · Project Jupyter · Docker/Moby · GnuCOBOL

Links and dataset attribution are listed in the source repository README.

Z-Mirror makes a decision traceable; it does not claim every release should pass.